

heyserial

heyserial - mandiant, GitHub.

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README.md

Blob `f83cd176ab6d`, 7393 bytes, at commit `78508ba5800b`.

HEY SERIAL!

Author: Alyssa Rahman @ramen0x3f

Created: 2021-10-27

Last Updated: 2021-12-02

Blog: <https://www.mandiant.com/resources/hunting-deserialization-exploits>

For more details on this tool and the research process behind it, check out [our blog](#)!

Description

Programmatically create hunting rules for deserialization exploitation with multiple

- keywords (e.g. cmd.exe)
- gadget chains (e.g. CommonsCollection)
- object types (e.g. ViewState, Java, Python Pickle, PHP)
- encodings (e.g. Base64, raw)
- rule types (e.g. Snort, Yara)

Disclaimer

Rules generated by this tool are intended for hunting/research purposes and are not designed for high fidelity/blocking purposes.

Please *test thoroughly* before deploying to any production systems.

The Yara rules are primarily intended for scanning web server logs. Some of the "object prefixes" are only 2 bytes long, so they can make large scans a bit slow. (*Translation: please don't drop them all into VT Retrohunt.*)

Usage

Help:

Examples:

```
python3 heyserial.py -c 'ExampleChain::condition1+condition2' -t JavaObj python3  
heyserial.py -k cmd.exe whoami 'This file cannot be run in DOS mode' python3  
heyserial.py -k Process.Start -t NETViewState -e base64 "base64+utf16le"
```

Utils

utils/checkyoself.py

This is a tool to automate bulk testing of Snort and Yara rules on a variety of sample files.

Usage:

```
```python3 checkyoself.py [-y rules.yara] [-s rules.snort] [-o file_output_prefix] [--matches] [--misses] -d malware.exe malware.pcap```
```

Examples:

```
```python3 checkyoself.py -y rules/javaobj -s rules/javaobj -d payloads/javaobj pcaps --misses -o java_misses```
```

utils/generate_payloads.ps1

YSoSerial.NET v1.34 payload generation. Run on Windows **from** the ./utils directory.

- Source: <https://github.com/pwntester/ysoserial.net>
- License: ysoserial.net_LICENSE.txt

utils/generate_payloads.sh

YSoSerial payload generation. Run on Linux **from** the ./utils directory.

- Source: <https://github.com/frohoff/ysoserial>
- License: ysoserial_LICENSE.txt

utils/install_snort.sh

Installing Snort on a Debian based system was a bit finnickty **for** me, so I wrote my install notes here.

Use at your own risk *in a VM* that *you have snapshotted recently*.

utils/server.py

Simple Python script that runs an HTTP server on 127.0.0.1:12345 and accepts POST requests.

Handy **for** generating test PCAPs.

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Contributing

Check out the Developers' guide (DEVELOPERS.md) [for](#) more details on extending HeySerial!

Prior Work/Related Resources

Tools

- [Deserialization-Cheat-Sheet](<https://github.com/GrrrDog/Java-Deserialization-Cheat-Sheet>) – @GrrrDog
- [Ysoserial](<https://github.com/frohoff/ysoserial>) - @frohoff
- [MarshalSec](<https://github.com/frohoff/marshalsec>) - @frohoff
- [Ysoserial (forked)](<https://github.com/wh1t3plg/ysoserial>) - @wh1t3plg
- [Ysoserial.NET](<https://github.com/pwntester/ysoserial.net>) and [v2 branch](<https://github.com/pwntester/ysoserial.net/tree/v2>) - @pwntester
- [ViewGen](<https://github.com/0xacb/viewgen>) – 0xacb
- [Rogue-JNDI](<https://github.com/veracode-research/rogue-jndi>) - @veracode-research

Vulnerabilities

- Log4J ([CVE-2021-44228](<https://www.lunasec.io/docs/blog/log4j-zero-day/>))
- Exchange ([CVE-2021-42321](<https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2021-42321>))
- Zoho ManageEngine ([CVE-2020-10189](<https://nvd.nist.gov/vuln/detail/CVE-2020-10189>))
- Jira ([CVE-2020-36239](<https://oxalis.io/atlassian-jira-data-centers-critical-vulnerability-what-you-need-to-know/>))
- Telerik ([CVE-2019-18935](<https://bishopfox.com/blog/cve-2019-18935-remote-code-execution-in-telerik-ui>))
- C1 CMS ([CVE-2019-18211](<https://medium.com/@frycos/yet-another-net-deserialization-35f6ce048df7>))
- Jenkins ([CVE-2016-9299](<https://nvd.nist.gov/vuln/detail/CVE-2016-9299>))
- [What [Do](#) WebLogic, WebSphere, JBoss, Jenkins, OpenNMS, and Your Application Have in Common? [This](#) Vulnerability.](<https://foxglovesecurity.com/2015/11/06/what-do-weblogic-websphere-jboss-jenkins-opennms-and-your-application-have-in-common-this-vulnerability/>) – @breenmachine, FoxGloveSecurity (2015)

Talks and Write-Ups

- [PSA: Log4Shell and the current state of JNDI injection](https://mbechler.github.io/2021/12/10/PSA_Log4Shell_JNDI_Injection/) - Moritz Bechler (2021)
- [[This](#) is Not a Test: APT41 Initiates Global Intrusion Campaign [Using](#) Multiple Exploits](<https://www.mandiant.com/resources/apt41-initiates-global-intrusion-campaign-using->

multiple-exploits) – Chris Glycer, Dan Perez, Sarah Jones, Steve Miller (2020)

- [Deep Dive into .NET ViewState deserialization and its exploitation] (<https://swapneildash.medium.com/deep-dive-into-net-viewstate-deserialization-and-its-exploitation-54bf5b788817>) – Swapneil Dash (2019)
- [Exploiting Deserialization in ASP.NET via ViewState] (<https://soroush.secproject.com/blog/2019/04/exploiting-deserialisation-in-asp-net-via-viewstate/>) – Soroush Dalili (2019)
- [Use of Deserialization in .NET Framework Methods and Classes] (<https://research.nccgroup.com/wp-content/uploads/2020/07/whitepaper-new.pdf>) – Soroush Dalili(2018)
- [Friday the 13th, JSON Attacks](<https://www.blackhat.com/docs/us-17/thursday/us-17-Munoz-Friday-The-13th-JSON-Attacks-wp.pdf>) – Alvaro Muñoz and Oleksandr Mirosh (2017)
- [Exploiting .NET Managed DCOM] (<https://googleprojectzero.blogspot.com/2017/04/exploiting-net-managed-dcom.html>) – James Forshaw, Project Zero (2017)
- [Java Unmarshaller Security] (<https://github.com/frohoff/marshalsec/blob/master/marshalsec.pdf>) – Moritz Bechler (2017)
- [Deserialize My Shorts](<https://www.slideshare.net/frohoff1/deserialize-my-shorts-or-how-i-learned-to-start-worrying-and-hate-java-object-deserialization>) – Chris Frohoff (2016)
- [Pwning Your Java Messaging with Deserialization Vulnerabilities] (<https://www.blackhat.com/docs/us-16/materials/us-16-Kaiser-Pwning-Your-Java-Messaging-With-Deserialization-Vulnerabilities-wp.pdf>) – Matthias Kaiser (2016)
- [Journey from JNDI/LDAP Manipulation to Remote Code Execution Dream Land] (<https://www.blackhat.com/docs/us-16/materials/us-16-Munoz-A-Journey-From-JNDI-LDAP-Manipulation-To-RCE-wp.pdf>) – Alvaro Muñoz and Oleksandr Mirosh (2016)
- [Marshalling Pickles](<https://www.youtube.com/watch?v=KSA7vUkXGSg>) – Chris Frohoff and Gabriel Lawrence (2015)
- [Are you my Type? Breaking .NET Through Serialization] (https://github.com/VulnerableGhost/.Net-Sterilized--Deserialization-Exploitation/blob/master/BH_US_12_Forshaw_Are_You_My_Type_WP.pdf) – James Forshaw (2012)
- [A Spirited Peek into ViewState](<https://deadliestwebattacks.com/2011/05/13/a-spirited-peek-into-viewstate-part-i/>) – Mike Shema (2011)